

A formal proof, in Lean 4, that one of $\zeta(5), \zeta(7), \dots, \zeta(33)$ is irrational

Formalization of W. Zudilin’s elementary argument
(arXiv:1801.09895), carried out in Lean 4 / Mathlib

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Abstract

We report a complete, machine-checked formalization, in the Lean 4 proof assistant on top of Mathlib, of a theorem of the shape “at least one of the odd zeta values $\zeta(5), \zeta(7), \dots, \zeta(s)$ is irrational.” The argument formalized is the elementary one of W. Zudilin [3], whose only analytic inputs are Stirling’s formula and a bound on the least common multiple $d_n = \text{lcm}(1, \dots, n)$. Because the current Mathlib does not contain the prime number theorem, we cannot use Zudilin’s headline value $s = 25$ (which needs $d_n^{1/n} \rightarrow e$); instead we formalize, from scratch and sorry-free, Hanson’s elementary bound $d_n \leq 3^n$ and take $s = 33$ — precisely the fallback that Zudilin himself records. The final theorem

$$\exists j \in \{5, 7, 9, \dots, 33\}, \quad \zeta(j) \notin \mathbb{Q},$$

is proved with no axioms beyond Lean’s three standard ones (`propext`, `Classical.choice`, `Quot.sound`); in particular it uses no `sorry` and no kernel-external computation (`native_decide`). We describe the construction, the four analytic and arithmetic ingredients, the reconstruction of Hanson’s bound, and the engineering of the development.

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1 Introduction

The irrationality of $\zeta(s) = \sum_{n \geq 1} n^{-s}$ is classical for even s and, after Apéry, for $s = 3$; for odd $s \geq 5$ only partial results are known. The prototypical such result — “infinitely many odd zeta values are irrational,” and its finite refinements “at least one of $\zeta(5), \dots, \zeta(s)$ is irrational” — is due to Ball–Rivoal [1] and Rivoal, and traditionally rests on the saddle-point method and/or linear-independence criteria, both regarded as decidedly non-elementary. Zudilin [3] gave, “by elementary means,” the first proof of such a statement whose only technical ingredients are the prime number theorem and Stirling’s approximation. The present work is a full formal verification of that argument.

The formalized statement

Let, in Lean,

$$\text{zetaVal } j = \sum_{n=0}^{\infty} \frac{1}{(n+1)^j} = \zeta(j), \quad \text{oddIdx} = \{j \in \mathbb{N} : 5 \leq j \leq 33, j \text{ odd}\}.$$

Our main theorem is

$$\text{zeta_odd_irrational} : \exists j \in \text{oddIdx}, \text{Irrational}(\text{zetaVal } j), \quad (1)$$

i.e. *at least one of $\zeta(5), \zeta(7), \dots, \zeta(33)$ is irrational.* Here `Irrational` is Mathlib’s predicate $x \notin \text{range}(\mathbb{Q} \hookrightarrow \mathbb{R})$. A `#print axioms` query returns

[propext, Classical.choice, Quot.sound]

and the whole development compiles with no `sorry`.

Why $s = 33$ and not $s = 25$

Zudilin’s optimal elementary result is for $s = 25$. Its proof uses the prime number theorem in the equivalent form $\lim_{n \rightarrow \infty} d_n^{1/n} = e$. Mathlib (at the commit used here, on `leanprover/lean4:v4.33.0-rc1`) does *not* contain the prime number theorem; its sharpest built-in bound is the Chebyshev estimate $\psi(x) \leq x \log 4 + o(x)$, giving only $\limsup_n d_n^{1/n} \leq 4$, which is too weak for *every* s . The external *PrimeNumberTheoremAnd* project would pin an incompatible Mathlib revision and cannot be added without a forbidden dependency update. We therefore follow Zudilin’s own remark ([3], §4): using the weaker, *elementary* bound $d_n < 3^n$ of Hanson [2] and the choice $s = 33$ “to arrive at the same conclusion.” Reconstructing Hanson’s bound in Lean turned out to be one of the substantial pieces of the project (§5).

2 The construction

Fix an odd $s \geq 7$ and write $q = (s+1)/2 \geq 4$; the formalization takes $s = 33$, $q = 17$, but all of the asymptotic analysis is developed for general q . Set

$$R_n(t) = \frac{n!^{s-5} 2^{6n} \prod_{j=1}^n (t-j) \prod_{j=1}^n (t+n+j) \prod_{j=1}^{3n} (t-n-\frac{1}{2}+j)}{\prod_{j=0}^n (t+j)^s}. \quad (2)$$

The two families of linear forms are

$$r_n = \sum_{\nu=1}^{\infty} R_n(\nu), \quad \hat{r}_n = \sum_{\nu=1}^{\infty} R_n(\nu - \tfrac{1}{2}).$$

Since R_n vanishes at $1, \dots, n$ and at $\frac{1}{2}, \dots, n - \frac{1}{2}$, both sums may be started at $\nu = n + 1$; the corresponding summands, written in factorial form, are the Lean definitions $\mathbf{c_q_n_k} = R_n(n+1+k)$ and $\mathbf{chat_q_n_k} = R_n(n + \frac{1}{2} + k)$, and

$$\mathbf{r_q_n} = \sum_{k \geq 0} \mathbf{c_q_n_k}, \quad \mathbf{rhat_q_n} = \sum_{k \geq 0} \mathbf{chat_q_n_k}.$$

The proof has four ingredients, mirrored by the module layout:

- (A) *Arithmetic* (§3, `Forms.lean`): r_n and $\hat{r}_n$ are $\mathbb{Z}[1/d_n]$ -linear combinations of $1, \zeta(3), \zeta(5), \dots, \zeta(s)$, and the “twist by half” makes their $\zeta(3)$ -coefficients proportional, so that $7r_n - \hat{r}_n$ lives in $\mathbb{Q} + \mathbb{Q}\zeta(5) + \dots + \mathbb{Q}\zeta(s)$.
- (B) *Asymptotics* (§4, Lemma 4): $r_n^{1/n}, \hat{r}_n^{1/n} \rightarrow g(x_0)$ and $r_n/\hat{r}_n \rightarrow 1$.
- (C) *The denominator bound* (§5, `DnBound.lean`): $d_n \leq 3^n$.
- (D) *The numerical estimate* (§6, `Numeric.lean`): $3^{33} g(x_0) < 1$.

Assembling these yields the contradiction (§7).

3 Arithmetic of the linear forms

This is the combinatorial heart of the development (`Zeta50dd/Forms.lean`). We reproduce Zudilin’s Lemmas 1–3 and the $\zeta(3)$ -elimination.

3.1 Partial fractions and integrality (Lemmas 1–2)

Write $R_n(t) = \sum_{i=1}^s \sum_{k=0}^n a_{i,k}/(t+k)^i$ (its partial-fraction decomposition, eq. (e04) of [3]). Two facts are needed.

Lemma 1 (Integrality; Lemma 1 of [3]). $d_n^{s-i} a_{i,k} \in \mathbb{Z}$ for $1 \leq i \leq s$, $0 \leq k \leq n$.

Lemma 2 (Well-poised symmetry; Lemma 2). $a_{i,k} = (-1)^{i-1} a_{i,n-k}$; hence $\sum_k a_{i,k} = 0$ for even i .

In Lean these are packaged as

`partialFraction_exists` $n : \exists a : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R}, (\text{decomposition}) \wedge (\text{integrality}) \wedge (\text{symmetry})$.

The three conjuncts are proved as follows.

Decomposition and integrality (`pf_decomp`) follow Zudilin’s route of writing R_n as a product of the six “simpler” rational functions of [3] (eqs. before (e04)), each of which has denominator $\prod_{j=0}^n (t+j)$ and *integer* partial-fraction coefficients given by explicit binomials, e.g.

$$\frac{n!}{\prod_{j=0}^n (t+j)} = \sum_{k=0}^n \frac{(-1)^k \binom{n}{k}}{t+k}, \quad \frac{2^{2n} \prod_{j=1}^n (t - \tfrac{1}{2} + j)}{\prod_{j=0}^n (t+j)} = \sum_{k=0}^n \frac{\binom{2k}{k} \binom{2n-2k}{n-k}}{t+k}.$$

Each of these six identities is proved (lemmas `f1_id`, ..., `f6_id`) via a reusable “clear the denominator and interpolate” engine `pf_general`, which reduces the claim to a polynomial identity of degree $< n+1$ holding at the $n+1$ nodes $t = -k$ (Mathlib’s `Polynomial.eq_of_natDegree_lt_card_of_eval_eq`). The product of 28 copies of the first simple function with the remaining five reproduces R_n (lemma `Rn_as_simpleProd`, matching the $n!^{28}$, 2^{6n} and denominator power $s = 33 = 28 + 5$). Integrality of the product’s coefficients is an induction on the number of factors (`pf_prod`, step `pf_mul_simple`) whose analytic engine is the two-pole identity

$$\frac{1}{u^i(u+\delta)} = \sum_{r < i} \frac{(-1)^r}{\delta^{r+1} u^{i-r}} + \frac{(-1)^i}{\delta^i(u+\delta)} \quad (\text{pf_two_pole}),$$

applied with $u = t + k$, $\delta = (k' - k)$; the δ^{-p} factors are cleared by d_n^p because $|k' - k| \leq n$ divides d_n (`dvd_lcmUpto`). Uniqueness of the coefficients (`pf_unique`), needed to transport the symmetry, is proved by a residue-peeling limit argument: near a pole $-k_0$ the scaled difference $(t + k_0)^P \sum (a - b)_{i,k} / (t + k)^i$ tends both to 0 and to $(a - b)_{P,k_0}$.

Symmetry (Lemma 2) is derived from the well-poised functional equation

$$R_n(-t - n) = -R_n(t) \quad (\text{Rn_wellPoised}),$$

proved directly from (2) by reindexing the four finite products (the global sign -1 coming from s being odd), combined with uniqueness.

3.2 The zeta representation (Lemma 3)

Lemma 3 (Lemma 3 of [3]). *With $a_i = \sum_k a_{i,k}$,*

$$r_n = \sum_{\substack{i=3 \\ i \text{ odd}}}^s a_i \zeta(i) + a_0, \quad \hat{r}_n = \sum_{\substack{i=3 \\ i \text{ odd}}}^s a_i (2^i - 1) \zeta(i) + \hat{a}_0,$$

with $d_n^{s-i} a_i \in \mathbb{Z}$ and $d_n^s a_0, d_n^s \hat{a}_0 \in \mathbb{Z}$.

The Lean statement `repr_combined` multiplies through by d_n^s and plumbs the integer constants through existentials. The r_n half (eq. (e07)) uses $\sum_\nu (\nu + k)^{-i} = \zeta(i) - \sum_{\ell \leq k} \ell^{-i}$ (`tsum_shift_zeta`); a `tsum`/finite-sum interchange after discarding the even columns (Lemma 2) and the $i = 1$ column ($\sum_k a_{1,k} = 0$, from convergence of the residue sum); and the harmonic integrality $d_n^i \sum_{\ell \leq k} \ell^{-i} \in \mathbb{Z}$ (`harmonic_integrality`), which is absorbed into a_0 . The $\hat{r}_n$ half (eq. (e08)) is the analytic subtlety of the paper: R_n does not vanish at the half-integer nodes, so the summation is shifted to start at $\nu = -m - \frac{1}{2}$ with $m = \lfloor (n-1)/2 \rfloor$ (`rhat_shift`); the resulting half-integer harmonic heads have denominators $\leq n$ and are cleared by d_n and d_{n-1} (`odd_harmonic_integrality`, `signed_harmonic_integrality`); the identity $\sum_{\ell \geq 1} (\ell - \frac{1}{2})^{-i} = (2^i - 1)\zeta(i)$ supplies the coefficient $(2^i - 1)$.

3.3 Elimination of $\zeta(3)$

Because the $\zeta(i)$ -coefficient of $\hat{r}_n$ is $a_i(2^i - 1)$, that of $7r_n - \hat{r}_n$ is $a_i(7 - (2^i - 1)) = a_i(8 - 2^i)$, which vanishes at $i = 3$ (as $2^3 - 1 = 7$). Hence, for $s = 33$,

$$\text{elim_integer } n : d_n^{33} (7r_n - \hat{r}_n) = \sum_{j \in \text{oddIdx}} A_j \zeta(j) + A_0, \quad A_j, A_0 \in \mathbb{Z}, \quad (3)$$

where $A_j = d_n^j a_j (8 - 2^j)$ and $A_0 = 7a_0 - \hat{a}_0$, both integral by Lemma 3. This is the exact interface consumed by the endgame.

4 Asymptotics (Lemma 4)

The asymptotic analysis (modules `Basic`, `Localize`, `Window`, `Roots`, `Ratio`) is developed for all $q \geq 4$. Writing

$$f(x) = \frac{x+3}{x} \left(\frac{x+1}{x+2} \right)^q, \quad g(x) = \frac{2^6(x+3)^6(x+1)^{s+1}}{(x+2)^{2(s+1)}},$$

and letting $x_0 \in (0, 1)$ be the unique positive solution of $f(x) = 1$, one has

$$\text{tendsto_root_r}, \text{tendsto_root_rhat} : r_n^{1/n}, \hat{r}_n^{1/n} \longrightarrow g(x_0), \quad \text{tendsto_ratio} : r_n/\hat{r}_n \rightarrow 1.$$

The proof localizes both series to a window $|k - x_0 n| \leq \varepsilon n$, controls the termwise ratio $c_{k+1}/c_k \sim f(k/n)^2$ via a quantitative Stirling estimate, and passes to n -th roots, all of which absorb the subexponential prefactors. This part predates the present campaign and is used as a black box; it supplies in particular

$$\text{seven_r_sub_rhat_pos_eventually} : \forall^\infty n, 7r_n - \hat{r}_n > 0.$$

5 The denominator bound $d_n \leq 3^n$

Since Mathlib lacks the prime number theorem, the bound $d_n \leq 3^n$ (module `Zeta50dd/DnBound.lean`) had to be built from the ground up. This is an elementary but genuinely intricate theorem of Hanson [2]; our formalization is, to our knowledge, new.

5.1 Divisibility

Let $a_0 = 2, a_1 = 3, a_2 = 7, a_3 = 43, \dots$ be the Sylvester sequence ($a_{k+1} = a_0 a_1 \cdots a_k + 1$; `sylv`, `sylvProd`) and set

$$C_n = \frac{n!}{\prod_{i \geq 0} (\lfloor n/a_i \rfloor!)} \quad (\text{hansonC}).$$

Only finitely many factors are non-trivial, since $a_i > n$ forces $\lfloor n/a_i \rfloor = 0$. One shows $C_n \in \mathbb{Z}$ (`hansonDenom_dvd`), and, crucially,

$$d_n = \text{lcm}(1, \dots, n) \mid C_n \quad (\text{lcmUpto_dvd_hansonC}),$$

by comparing p -adic valuations through Legendre's formula and the arithmetic inequality $\sum_i \lfloor m/a_i \rfloor \leq m - 1$ (`core\_sum\_le`), itself a consequence of $\sum_i a_i^{-1} = 1 - 1/(a_0 \cdots a_{N-1}) < 1$ (`sum\_inv\_sylv`).

5.2 Size

It remains to bound $C_n \leq 3^n$, equivalently $\log n! \leq n \log 3 + \sum_i \log \lfloor n/a_i \rfloor!$. The margin is thin: the limiting exponential rate is $\sum_i (\log a_i)/a_i = 1.08239 \dots < \log 3 = 1.09861 \dots$, a gap of only 0.0162 per unit n . The Lean proof splits at $N_0 = 1600$:

- for $n < N_0$ the bound is reduced (via the truncation $C_n = n! / \prod_{i < 4} \lfloor n/a_i \rfloor!$, valid for $n < a_4 = 1807$) to a finite $\mathbb{N}$ -inequality and discharged by the Lean kernel's `decide` (GMP arithmetic; `no native_decide`, hence no extra axiom);
- for $n \geq N_0$ we combine sharp two-sided Stirling bounds (`log_factorial_le/ge`), the tangent line of the convex $t \mapsto t \log t$, and a *certified rational* bound $\sum_i (\log a_i)/a_i < \log 3$ (obtained from integer inequalities such as $7^{26} < 2^{73}$, $43^7 < 2^{38}$ and a geometric tail), through a self-bootstrapping induction on the Sylvester index (`hanson_key`).

The whole file certifies `lcmUpto_le_three_pow` : $(d_n : \mathbb{R}) \leq 3^n$ with axioms `[propext, Classical.choice, Quot.sound]`.

6 The numerical estimate

At x_0 one has $x_0 = 2.2893\dots \times 10^{-5}$ and $\log g(x_0) = -36.3834\dots$, while $3^{33} = e^{36.2543\dots}$, so $3^{33}g(x_0) = e^{-0.129} < 1$. The Lean lemma `g_small` proves $3^{33}g(17, x) < 1$ for the (unique) positive x with $f(17, x) = 1$. Two ingredients: (i) $x \leq \frac{1}{40}$, by a quadratic–Bernoulli argument reducing $f(x) = 1$ to a factored polynomial inequality; and (ii) g increasing near 0 (via the sign of $(\log g)'$), so $g(x) \leq g(\frac{1}{40})$, together with the *exact rational* inequality

$$3^{33}g(17, \frac{1}{40}) < 1 \iff 64 \cdot 121^6 \cdot 41^{34} \cdot 40^{28} < 3^{239},$$

closed by `norm_num`.

7 Assembly and the contradiction

The endgame (`Zeta50dd/Main.lean`) is Zudilin’s final page. Suppose, for contradiction, that every $\zeta(j)$ with $j \in \text{oddIdx}$ is rational; a common denominator $a \in \mathbb{N}_{>0}$ then satisfies $a\zeta(j) \in \mathbb{Z}$ (`exists_common_denom`). Put $b_n = d_n^{33}(7r_n - \hat{r}_n)$. From (3),

$$a b_n = \sum_{j \in \text{oddIdx}} A_j (a\zeta(j)) + a A_0 \in \mathbb{Z}.$$

On the other hand $b_n \geq 0$ and, by §4–6, the n -th root of b_n satisfies

$$b_n^{1/n} = (d_n^{1/n})^{33} (7r_n - \hat{r}_n)^{1/n} \leq 3^{33} (7r_n - \hat{r}_n)^{1/n} \longrightarrow 3^{33}g(x_0) < 1,$$

using $d_n \leq 3^n$ and $(7r_n - \hat{r}_n)^{1/n} \rightarrow g(x_0)$ (`tendsto_seven_root`, deduced from Lemma 4). Hence a root test gives $b_n \rightarrow 0$, so $a b_n \rightarrow 0$; but $7r_n - \hat{r}_n > 0$ eventually makes $a b_n$ a *positive integer*, so $a b_n \geq 1$ — a contradiction. This is (1).

8 Verification and engineering

Toolchain. Lean v4.33.0-rc1, Mathlib at commit cd580e54 (master). The development is one Lake library, `Zeta50dd`, of ten modules.

Trust. The final theorem `zeta_odd_irrational` depends only on `[propext, Classical.choice, Quot.sound]`. There is no `sorry` anywhere in the library, and no use of `native_decide` (which would add `Lean.ofReduceBool`); the one finite case analysis (§5) uses kernel `decide`.

Module map.

<code>Basic</code> , <code>Localize</code> , <code>Window</code> , <code>Roots</code> , <code>Ratio</code>	Lemma 4 asymptotics.
<code>ZetaValues</code>	ζ -values, <code>oddIdx</code> , common denominator.
<code>Forms</code>	Lemmas 1–3, well-poised symmetry, $\zeta(3)$ -elimination.
<code>DnBound</code>	Hanson’s $d_n \leq 3^n$.
<code>Numeric</code>	$3^{33}g(x_0) < 1$.
<code>Main</code>	root-test endgame; theorem (1).

Method. The proof was assembled by an orchestration of Opus language-model agents: the inter-module interface statements were fixed in advance (so that the arithmetic, asymptotic, denominator and numeric parts could be developed against stable targets), and the pieces were proved in roughly a dozen focused rounds and merged, keeping the build green throughout.

Remark 4 (On sharpness). Nothing here improves on Zudilin; on the contrary, formalizing $d_n \leq 3^n$ rather than the prime number theorem costs the eight extra zeta values $\zeta(27), \dots, \zeta(33)$. Once Mathlib gains the prime number theorem, the single lemma `lcmUpto_le_three_pow` can be replaced by $d_n^{1/n} \rightarrow e$ and the constant 33 throughout `Forms`, `Numeric`, `Main` lowered to 25, recovering Zudilin’s optimal statement with no change to the arithmetic or asymptotic machinery.

References

- [1] K. Ball, T. Rivoal, *Irrationalité d’une infinité de valeurs de la fonction zêta aux entiers impairs*, Invent. Math. **146** (2001), 193–207.
- [2] D. Hanson, *On the product of the primes*, Canad. Math. Bull. **15** (1972), 33–37.
- [3] W. Zudilin, *One of the odd zeta values from $\zeta(5)$ to $\zeta(25)$ is irrational. By elementary means*, SIGMA **14** (2018), 028, arXiv:1801.09895.